Finding Belief Geometries with Sparse Autoencoders

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Abstract

Transformers trained on naturalistic sequences may internally represent the distributions over latent generative states—so-called *belief states*—as structured geometric objects. Prior work has shown that the residual stream of transformers trained on data from hidden Markov models exhibits simplex-shaped representations aligned with the true belief geometry of the underlying process. Whether large language models encode analogous structures in their internal representations has remained an open question. We introduce a pipeline that combines sparse autoencoders (SAEs), k-subspace latent clustering, and a simplex-fitting autoencoder (AANet) to search for simplex-structured subspaces in the SAE latent space of Gemma-2-9B. As a proof of concept we first validate the approach on a controlled toy model—a small transformer trained on a multipartite hidden Markov model with known ground-truth belief geometries—and demonstrate that the pipeline can recover the expected simplex structures. Applying the pipeline to Gemma-2-9B, we identify 13 priority clusters exhibiting 2-simplex ($K=3$) structure. We evaluate these clusters across four validation strands: (1) cross-vertex KL divergence of next-token distributions, (2) the predictive advantage of barycentric over individual latent coordinates, (3) causal steering of model continuations via AANet-guided activations, and (4) semantic coherence of vertex labels produced by a frontier language model. For 3 of the 13 clusters, barycentric simplex coordinates outperform the best individual latent at predicting next-token distributions (Wilcoxon $p < 10^{-14}$), an advantage not observed in size-matched null clusters. Causal interventions guided by simplex coordinates shift model continuations toward target vertex semantics in 8 of 13 clusters, with the strongest cluster achieving a mean shift rate of 0.42. Together, the evidence is consistent with—though does not conclusively prove—the hypothesis that Gemma-2-9B encodes latent-generative-state mixtures as functional simplex geometries in subspaces of its SAE latent representation.

1 Introduction

A central challenge in mechanistic interpretability is to characterize the *representational geometry* used by language models to encode the context that drives their predictions. One principled lens on this question comes from the theory of predictive processing: a statistically optimal sequence predictor must maintain an internal representation of its belief over the latent state of the generative process—a distribution over hidden states that is updated by each new observation [Marzen and Crutchfield, 2017, Shai et al., 2024]. For generative processes with finite latent-state spaces (e.g., hidden Markov models), the space of reachable belief states forms a convex polytope—a simplex when the hidden states are three in number—and the optimal internal representation should reflect this geometry [Shai et al., 2024, Piotrowski et al., 2025].

Empirical evidence supports this picture: transformers trained on data from hidden Markov models embed the true simplex geometry in their residual stream, recoverable by PCA and coloring by ground-truth hidden state [Shai et al., 2024]. Moreover, the specific attention mechanisms that implement belief updating are theoretically predictable from the structure of the data-generating process [Piotrowski et al., 2025], and the same phenomenon extends to neural networks learning over quantum and post-quantum generative models [Riechers et al., 2025b]. Separately, Riechers et al. [2025a] have shown that successful next-token pretraining mathematically implies in-context learning, with belief-state synchronization as the underlying mechanism—a result that motivates identifying the specific representational subspaces where this synchronization occurs. Whether the same phenomenon appears in large-scale language models (LLMs) trained on naturalistic text is less clear. The generative processes underlying natural language are vastly more complex, involving many interacting latent factors (topic, register, syntactic context, referential state, etc.), and the simplex structures—if present—would presumably be encoded in distributed, high-dimensional subspaces rather than in the leading principal components of the full residual stream.

Sparse autoencoders (SAEs) offer a potential bridge: by learning a sparse overcomplete representation of residual stream activations, they decompose the representational space into interpretable features [Bricken et al., 2023, Cunningham et al., 2023]. If a language model encodes a latent state mixture via a simplex, we would expect a corresponding cluster of co-active SAE latents whose decoder directions together span a subspace approximating that simplex. Discovering such clusters and verifying their geometric and functional properties constitutes the central aim of this work.

We make the following contributions:

1. We introduce a discovery pipeline combining k-subspace clustering of SAE decoder directions with AANet [Korem et al., 2019] simplex fitting to identify candidate simplex-structured subspaces.
2. We validate the pipeline end-to-end on a controlled toy model with known ground-truth belief geometries.
3. Applying the pipeline to Gemma-2-9B with a publicly available SAE, we identify 13 priority clusters exhibiting $K=3$ simplex structure.
4. We characterize these clusters with four complementary validation measurements: KL divergence, latent-vs.-barycentric predictive comparison, causal steering, and frontier-LLM interpretability.

Our findings are best summarized as *consistent with but not proving* the hypothesis that Gemma-2-9B encodes components of latent generative-state mixtures as functional simplex geometries. We discuss both positive evidence and honest limitations throughout.

2 Background and Related Work

Belief states and simplex geometry. For a stationary process generated by a hidden Markov model (HMM) with K hidden states, the belief state at time t is the posterior distribution $\mathbf{b}_t \in \Delta^{K-1}$ (the $(K-1)$-simplex). The Baum–Welch equations give the recurrence; in steady state the reachable belief states form the *belief simplex*, whose vertices correspond to certainty about a single hidden state. Shai et al. [2024] showed that transformers trained on such processes represent the belief simplex in their residual stream, with PCA projections clearly revealing the triangular or higher polytope structure, colored by the ground-truth belief. Piotrowski et al. [2025] further showed that the specific computational mechanism—constrained Bayesian belief updating implemented via attention—is theoretically predictable from the generative structure, and Riechers et al. [2025a] established that belief-state synchronization is the mechanism underlying in-context learning more broadly.

Sparse autoencoders. SAEs decompose residual stream activations into sparse sums of learned feature directions, with the goal of recovering monosemantic, interpretable features [Bricken et al., 2023, Cunningham et al., 2023]. We use the SAELens library [Bloom et al., 2024] and a publicly available TopK SAE for Gemma-2-9B (residual stream after layer 20, dimension 3584) with $d_{\text{SAE}} = 131072$ latents, sparsity level $K = 50$.

Archetypal analysis and AANet. Archetypal analysis [Cutler and Breiman, 1994] fits a $(K-1)$-simplex to a point cloud by finding K *archetypes* such that (i) the points can be approximated as convex combinations of the archetypes, and (ii) the archetypes themselves are convex combinations of extreme points. AANet [Korem et al., 2019] is a neural implementation that learns a nonlinear encoder mapping the input to barycentric coordinates and a nonlinear decoder that reconstructs the input from those coordinates. We use AANet to fit simplices to the activations of SAE latent clusters.

k-subspace clustering. We cluster SAE latent decoder directions using a variant of k-subspace clustering [Bradley and Mangasarian, 2000] initialized with QR-column-pivoting to select maximally orthogonal seed directions, which we motivate as finding near-orthogonal subspaces that may represent independent generative components.

Geometry of Concepts. Concurrent and related work [Chua et al., 2024] has studied the geometric structure of SAE features at a higher level of abstraction, finding that features cluster into structured polytopes. Our work is complementary: rather than studying the geometry of feature clusters from the SAE perspective, we use the SAE as a lens to look for belief-state geometry of potential latent generative components.

3 Toy Model: Proof of Concept

3.1 Generator Model

We construct a multipartite (MP) process by independently drawing emissions from five component models: three Mess3 hidden Markov models and two Tom Quantum (Bloch Walk) generalized HMMs [Riechers et al., 2025b]. The three Mess3 instances have parameters $(x, a) \in \{(0.05, 0.85), (0.075, 0.90), (0.10, 0.95)\}$; the two Tom Quantum instances have parameters $(\alpha, \beta) \in \{(1.51, 3.07), (1.99, 2.51)\}$. The MP model emission at each step is the Cartesian product of the five component emissions. With a vocabulary of $4 \times 4 \times 3 \times 3 \times 3 = 432$ tokens, the full token space is large enough to exercise interesting predictive structure while remaining tractable.

We fit a small TransformerLens transformer to predict next tokens from this MP generator. The architecture has $d_{\text{model}} = 128$, 4 heads, 3 layers, context length $n_{\text{ctx}} = 16$, and ReLU activations (total: 709k parameters). Training uses AdamW with base learning rate 10^{-4}, cosine decay to 10%, and sequences of length 16 in batches of 2048. Given the high conditional entropy of the process (the next-token distribution is nearly uniform), the model achieves modest top-1 accuracy of 7.1% (vs. 0.2% random) but learns meaningful representations of the belief geometries of the component processes.

3.2 Simplex Structure in the Residual Stream

Figure 1 shows PCA projections of the layer-2 residual stream, colored by ground-truth belief state, at sequence positions 5, 10, and 15. The three Mess3 components exhibit clear triangular (2-simplex) structure, becoming more diffuse but still recognizable as the sequence length grows. The two Tom Quantum components exhibit the expected circular (Bloch sphere projection) structure. These visualizations confirm that the toy transformer has learned to internally track the belief states of all five generator components and that this information is linearly readable from the residual stream.

3.3 SAE Fitting and Latent Analysis

We fit TopK SAEs with $K \in \{3, 4, 5, 6, 7\}$ and vanilla L1 SAEs with $\lambda \in \{10^{-3}, \ldots, 0.15\}$ to each layer of the toy model. The SAE latent space has dimension $d_{\text{SAE}} = 256$ for all variants. Figure 2 shows the empirical probability density functions (EPDFs) for the most active latents in the $K = 3$ TopK SAE at layer 2. Individual latents exhibit clear selectivity for specific regions of the belief simplex: for the Mess3 components, the top-8 latents tile the simplex with partially overlapping activation profiles, and each latent fires preferentially when the belief is concentrated near one vertex or edge.

3.4 Clustering and AANet Fitting

We apply k-subspace clustering to the toy SAE decoder matrix (256×128 for the layer-2, $K = 3$ SAE) to partition latents into N clusters. AANet is then fit to the cluster latent activations with target simplex dimensions $K \in \{2, 3, 4, 5\}$. We select K using an elbow criterion on the AANet reconstruction loss. For the Mess3-associated clusters, the reconstruction loss consistently reaches its inflection point at $K = 3$, correctly identifying the underlying 2-simplex.

The recovery is directionally correct but imperfect: the reconstructed simplex vertices are approximately aligned with the true belief-state corners, but noise in the SAE representation causes diffuse scatter and occasional vertex merging. This imperfection in the controlled setting is an important baseline for interpreting results on the real model (Section 4), where ground truth is unavailable and the representational noise is expected to be higher.

4 Real Model Pipeline: Gemma-2-9B

4.1 SAE and Representation Space

We use a publicly released TopK SAE ($K = 50$) for Gemma-2-9B trained on the residual stream after layer 20 [Bloom et al., 2024], with $d_{\text{model}} = 3584$ and $d_{\text{SAE}} = 131{,}072$ latents. Layer 20 corresponds to roughly 60% of the network depth, a range empirically associated with abstract contextual representations in similar models [Ferrando et al., 2024]. We process 5000 sequences from the Pile (context length 256) through Gemma-2-9B and store per-token SAE activations.

4.2 k-Subspace Clustering

The SAE decoder matrix $W_d \in \mathbb{R}^{3584\times 131072}$ is too large to cluster in its entirety. We restrict to the N SAE latents with the highest mean activation frequency across the corpus, forming a candidate pool. We cluster the corresponding column-normalized decoder directions using k-subspace clustering with QR-pivoting initialization. This yields a partition of decoder directions into groups that span approximately orthogonal subspaces of the residual stream, with the expectation that latents in the same cluster co-activate because they jointly encode a single underlying contextual variable.

We run the pipeline for $N \in \{512, 768\}$ (the top-N most active latents by frequency), producing $\lfloor\sqrt{N}\rfloor$ clusters per run and selecting candidate clusters with estimated rank $k = 3$ (i.e., evidence for 2-simplex structure from the cluster subspace elbow plot). We additionally apply spectral clustering as an independent check and retain only clusters identified by both methods.

4.3 AANet Simplex Fitting

For each candidate cluster of size m (number of latents), we form the cluster activation matrix $\mathbf{A} \in \mathbb{R}^{T\times m}$ where T is the number of token positions with at least one latent active. We project each activation vector onto the cluster decoder subspace: $\mathbf{x}_t = \mathbf{A}_t\mathbf{W}_c$, where $\mathbf{W}_c \in \mathbb{R}^{m\times d_{\text{model}}}$ is the cluster decoder submatrix. AANet is trained on $\{\mathbf{x}_t\}$ to minimize reconstruction loss subject to simplex constraints, producing barycentric coordinates $\mathbf{b}_t \in \Delta^{K-1}$.

The simplex dimension K is selected by fitting AANet for $K \in \{2, 3, 4, 5\}$ and identifying the elbow in the reconstruction loss curve. We focus on clusters where the elbow is at $K = 3$ (2-simplex), since this is the smallest non-degenerate simplex and the easiest to visualize and validate.

4.4 Priority Cluster Selection

After clustering and AANet fitting across both $N = 512$ and $N = 768$ runs, we select 13 priority clusters for validation (Table 1). All 13 have estimated $K = 3$ from the AANet elbow criterion. Cluster sizes range from 4 to 26 latents. For comparison, we also select three *null clusters*—random subsets of 47–51 latents drawn without any geometric selection criterion—and apply the identical AANet fitting and validation pipeline. Null clusters serve as a reference for the base rate of any measured signal under the null hypothesis of no genuine simplex structure.

5 Validation

We evaluate the 13 priority clusters across four complementary validation strands. None of the individual strands is conclusive in isolation; our argument rests on their collective pattern.

5.1 Strand 1: Cross-Vertex KL Divergence

Setup. For each cluster, we collect near-vertex samples: token positions where the AANet barycentric coordinate b_k for one vertex exceeds a threshold (0.70). For each pair of near-vertex samples (i, j), we run a forward pass to obtain the model's next-token distribution $P_i, P_j \in \Delta^{V-1}$ (vocabulary size $V = 256{,}000$) and compute the symmetric KL divergence $D_{\text{KL}}(P_i \| P_j)$. We compute three population statistics: *same-vertex*

Table 1: **The 13 priority clusters.** Clusters are identified by the number of top-frequency SAE latents used (512 or 768) and their cluster index within that run. Confidence reflects manual review of interpretability outputs and validation consistency. m denotes the number of SAE latents in the cluster.

Cluster	m	Confidence	Vertex interpretation (summary)
512_17	15	HIGH	Text register (instructional / promotional / metadata)
512_181	26	HIGH	[Pending: see main text]
768_140	26	HIGH	Referential specificity (generic / specific / anaphoric)
768_596	6	HIGH	Grammatical person / discourse role
512_22	6	MEDIUM	[Pending semantic label]
512_67	25	MEDIUM	[Pending semantic label]
512_229	22	MEDIUM	[Pending semantic label]
512_261	18	MEDIUM	[Pending semantic label]
512_471	10	MEDIUM	[Pending semantic label]
512_504	22	MEDIUM	[Pending semantic label]
768_210	4	MEDIUM	[Pending semantic label]
768_306	5	MEDIUM	[Pending semantic label]
768_581	11	MEDIUM	[Pending semantic label]

(both samples near the same vertex), *cross-vertex* (samples near different vertices), and *within-simplex* (randomly sampled active positions from the cluster, regardless of vertex proximity).

Hypothesis. If the simplex geometry is functionally meaningful—i.e., if different vertex positions correspond to genuinely different generative modes—then cross-vertex pairs should have larger next-token KL divergence than same-vertex pairs: $\overline{D}_{\mathrm{KL}}^{\mathrm{cross}} > \overline{D}_{\mathrm{KL}}^{\mathrm{same}}$.

Results. Table 2 shows that all 13 real clusters achieve cross/same KL ratios > 1, ranging from 1.017 to 1.987. However, the three null clusters also exhibit ratios > 1 (1.108–1.329), sometimes exceeding those of real clusters. The KL divergence test thus does not cleanly discriminate real from null clusters. One cluster stands out markedly: **768_140** achieves a ratio of 1.987, far higher than any other cluster (real or null), with a z-score of > 4 for the cross-vs.-same comparison. This result is consistent with 768_140's strongly interpretable semantic meaning (Section 6.2).

5.2 Strand 2: Latent vs. Barycentric Predictive Advantage

Setup. The most informative test of whether the simplex geometry captures meaningful information—beyond what any individual latent captures—is to ask whether the full barycentric coordinate vector $\mathbf{b}_t \in \mathbb{R}^K$ predicts the next-token distribution better than the single best individual latent in the cluster. For each of the top-50 most frequent tokens in the near-vertex sample, we fit a linear regression R^2 for the probability of that token given the barycentric coordinate (across $T \sim 180$ samples) and separately for the activation of each latent in the cluster. We compare R^2_{bary} vs. $R^2_{\mathrm{best\,latent}}$ using a paired Wilcoxon signed-rank test across the 50 tokens.

Hypothesis. If the simplex representation captures genuine mixture information not present in any single latent, then R^2_{bary} should be systematically larger than $R^2_{\mathrm{best\,latent}}$.

Results. Table 3 shows that 3 of the 13 real clusters (512_181, 512_229, 768_596) exhibit a significant barycentric advantage, with barycentric R^2 exceeding the best individual latent's R^2 for nearly all or all top-50 tokens, with Wilcoxon $p < 10^{-14}$. Crucially, none of the three null clusters exhibits a significant advantage ($p \approx 1.0$ in all cases). This is the single cleanest discriminating result in our validation battery: it directly measures whether the *mixture* information encoded in the full simplex coordinate carries independent

Table 2: **KL divergence results: cross-vertex vs. same-vertex ratio.** We report the ratio $\overline{D}_{\mathrm{KL}}^{\mathrm{cross}}/\overline{D}_{\mathrm{KL}}^{\mathrm{same}}$ of mean symmetric KL divergences. A ratio > 1 indicates that tokens near different simplex vertices predict more divergent next-token distributions than tokens near the same vertex. Null clusters are shown separately; their ratios overlap substantially with those of real clusters.

Cluster	m	KL ratio	Cluster	m	KL ratio
512_17	15	1.054	512_471	10	1.017
512_22	6	1.058	512_504	22	1.164
512_67	25	1.229	768_140	26	**1.987**
512_181	26	1.037	768_210	4	1.017
512_229	22	1.166	768_306	5	1.102
512_261	18	1.032	768_581	11	1.144
			768_596	6	1.037
Null clusters:					
512_138 (null)	47	1.329	512_345 (null)	51	1.311
768_310 (null)	48	1.108			

predictive signal, which is precisely what one would expect if the simplex geometry reflects a latent state variable over which the model is computing something like Bayesian mixture inference.

The improvement is substantial for 512_181 and 512_229 (mean R^2 improves by $\sim$14% and $\sim$55%, respectively) but more modest for 768_596 (improving by $\sim$24%). Conversely, 768_140—our most interpretable cluster—does not show a significant advantage despite its strong KL divergence result; here the best single latent ($R^2 = 0.549$) slightly exceeds the barycentric R^2 (0.528).

5.3 Strand 3: Causal Steering

Setup. We test whether the simplex geometry identified by AANet is *causally* relevant by steering the model's residual stream to target vertex coordinates and asking whether model continuations shift to semantically match the target vertex. Concretely, for each source vertex v (from which we start near-vertex samples) and each target vertex $v' \neq v$, we compute an intervention direction in residual stream space derived from the AANet vertex decoder, add it at scale $s \in \{1, 5, 20\}$ to the residual stream at the trigger position, and generate a continuation. We evaluate each continuation using Qwen-2.5-72B (AWQ quantized, via vLLM), which classifies whether the continuation matches the target vertex's semantic description (derived from the autointerp labels; Section 5.4). The steering score is the mean classification accuracy across source–target pairs, types (3 intervention formulations), and scales.

A cluster *qualifies* for steering analysis if at least 2 of its 3 vertices achieve document accuracy ≥ 0.15 (the Qwen classifier can reliably identify natural continuation text for those vertices), ensuring the score is not trivially driven by a degenerate vertex. Five real clusters do not qualify; we report the 8 that do.

Results. Table 4 shows that 8 of 13 real clusters qualify and achieve positive steering scores. The strongest cluster, 768_596, achieves a score of 0.419. However, the null cluster 512_138 achieves 0.358, and 768_310 achieves 0.272, overlapping with the lower half of the real cluster distribution. Causal steering thus does not cleanly discriminate real from null clusters.

Examining the null clusters more carefully reveals a consistent pattern: all three null clusters contain a *phantom vertex* with document accuracy < 0.15 (i.e., natural continuation text from that vertex cannot be reliably classified into it). High steering scores in these cases reflect not genuine geometric causal influence but rather that samples labeled as originating from the phantom vertex are misclassified in the baseline, so steering away from it yields apparent improvement. We note, however, that 768_596 also has a poorly defined third vertex (V2 document accuracy = 0.069), meaning this artifact is not unique to null clusters; 768_596's high score should be interpreted with this caveat in mind.

Despite the overlap with null clusters, the *combination* of strong steering and barycentric advantage for 768_596 is more compelling than either metric alone: the barycentric advantage is not present in any null

Table 3: **Barycentric vs. best-latent predictive advantage.** Fraction of top-50 tokens for which the barycentric coordinate achieves higher R^2 for next-token probability than the best individual latent in the cluster (*frac. bary wins*). Wilcoxon p-values test the one-sided hypothesis that barycentric R^2 is stochastically greater. Mean R^2 values are shown for context. Null clusters are shown at bottom; none achieves significant barycentric advantage.

Cluster	Frac. bary wins	Wilcoxon p	Mean R^2_{bary}	Mean $R^2_{\text{best latent}}$
512_17	0.00	1.00	0.055	0.138
512_22	0.00	—	—	—
512_67	0.00	—	—	—
512_181	**1.00**	$< 10^{-15}$	**0.612**	0.539
512_229	**1.00**	$< 10^{-15}$	**0.378**	0.244
512_261	0.00	—	—	—
512_471	0.00	—	—	—
512_504	0.00	—	—	—
768_140	0.20	≈ 1.0	0.528	**0.549**
768_210	0.00	—	—	—
768_306	0.16	≈ 1.0	—	—
768_581	0.00	—	—	—
768_596	**0.98**	$< 10^{-14}$	**0.269**	0.217
512_138 (null)	0.00	1.00	—	—
512_345 (null)	0.02	≈ 1.0	—	—
768_310 (null)	0.16	≈ 1.0	—	—

cluster, while the steering score for 768_596 is the highest across all clusters.

5.4 Strand 4: Semantic Coherence and LLM Interpretability

Setup. For each cluster, we collect the top-50 most activating tokens per vertex (near-vertex samples, sorted by activation weight) and submit them for interpretability to Claude Sonnet, repeating the labeling 20 times with different random seeds. We synthesize the vertex labels across iterations into a single consolidated description and assess inter-iteration consistency.

Results. All 13 priority clusters produce semantically coherent vertex labels that are stable across the majority of iterations. The three best-characterized clusters are described in detail in Section 6. Semantic coherence is necessary but not sufficient evidence: the null clusters also produce coherent vertex labels (as expected, since any set of tokens with a common clustering criterion will have some interpretable theme). We treat this strand primarily as a *prerequisite check*—if we cannot produce stable, coherent labels, the other quantitative tests lose interpretive grounding—rather than as an independent discriminating measurement.

6 Highlight Clusters

We profile three clusters that together illustrate the range of evidence available and that, in combination, offer the most compelling case for genuine simplex structure in Gemma-2-9B's representations.

6.1 768_596: Strongest Functional Evidence

Cluster 768_596 ($m = 6$ latents) is the single cluster with the strongest convergence of functional evidence: it achieves the highest steering score (0.419), and its barycentric coordinates significantly outperform the best individual latent at next-token prediction (frac. bary wins = 0.98, $p < 10^{-14}$). Figure 4 shows the latent spatial structure: the 5000 samples occupy a triangular region with concentration near the three vertices, and the 6 latents partition into three vertex-preferring groups.

Table 4: **Causal steering scores.** The steering score reports the best (type, scale) combination's mean shift rate across all source→target directions. Clusters where AANet did not qualify (fewer than 2 vertices with document accuracy ≥ 0.15) are marked with a dash. Null cluster scores are shown at bottom for comparison. Distributions of real and null scores overlap, but the best real cluster (768_596) exceeds all null clusters.

Cluster	Steering score	Best type	Best scale
768_596	**0.419**	type2	20
512_22	0.356	type1	20
768_210	0.336	type2	1
768_140	0.289	type1	20
512_67	0.250	type1	20
512_261	0.242	type1	1
768_581	0.209	type3	5
512_17	0.118	type3	5
512_181	—	—	—
512_229	—	—	—
512_471	—	—	—
512_504	—	—	—
768_306	—	—	—
512_138 (null)	0.358	type1	1
768_310 (null)	0.272	type1	20
512_345 (null)	0.150	type1	5

The semantic interpretation of this cluster is moderate-confidence. Claude Sonnet produces stable labels across 20 iterations: V0 corresponds to third-person or external-entity reference, V1 to first-person or self-referential framing, and V2 to second-person or direct-address text. However, the 6-latent cluster is small, and the trigger token distributions at each vertex are heterogeneous, including punctuation, function words, and content words, suggesting the true organizing principle may be closer to *syntactic discourse position* or *token type at discourse boundaries* than clean grammatical person.

6.2 768_140: Strongest KL Divergence and Best Interpretation

Cluster 768_140 ($m = 26$ latents) achieves a KL divergence ratio of 1.987, the largest in our dataset and far above both the other real clusters and the null clusters (Figure 5b). It also achieves the second-highest document classification accuracy at vertex 1 (0.958), indicating that natural text near V1 is highly classifiable.

The semantic interpretation of 768_140 is the clearest of any cluster in the dataset. Claude Sonnet consistently assigns:

- **V0**: Generic or indefinite reference—common nouns, abstract concepts, "a company", "some idea", "any kind of"
- **V1**: Specific or definite reference—proper nouns, named entities, dates, "The Apple Inc.", "Paris", "October 12"
- **V2**: Anaphoric or contextual reference—pronouns, demonstratives, "it", "they", "this phenomenon"

This three-way partition corresponds to the classical linguistic distinction between generic, specific, and referential noun phrase types, and maps naturally onto a latent-state interpretation: the model may be tracking *what kind of referential context it is in*, which strongly predicts upcoming token type (common noun vs. proper noun vs. pronoun), explaining the large KL signal.

Importantly, 768_140 does *not* exhibit a significant barycentric advantage ($p \approx 1.0$; Table 3): the best single latent outperforms the full barycentric coordinate by a small margin. This is consistent with a situation where the three reference types are sufficiently discrete that a single high-quality latent (specialized for "proper noun" or "anaphoric" reference) already captures most of the predictive signal, leaving little room for the mixture coordinate to add information.

6.3 512_17: Most Interpretable Semantic Structure

Cluster 512_17 ($m = 15$ latents, HIGH confidence) achieves the clearest and most consistent semantic interpretation of any cluster. Claude Sonnet assigns, with high stability across 20 iterations:

- **V0** [consistent]: Instructional/procedural text—how-to guides, technical step descriptions, user-action sequences
- **V1** [consistent]: Descriptive/promotional text—service descriptions, product features, business offerings
- **V2** [mixed]: Metadata/navigational/organizational context—company information, navigation elements, structural text

This partition corresponds to *document register* or text genre: the model may be tracking what type of discourse mode it is in, which governs the expected vocabulary and syntax of subsequent text. The belief-state interpretation is natural: the "generative process" is the human author, and the hidden state is the register they are writing in.

Quantitatively, 512_17 is one of the weaker clusters: its steering score (0.118) is the lowest among qualified clusters, and it shows no significant barycentric advantage. This may reflect that register is more distributed across many SAE features and that the 15-latent cluster captures only a portion of the relevant representational subspace, making causal interventions from it relatively weak.

7 Discussion

7.1 Convergent Evidence Without a Single Decisive Test

No single validation measure cleanly separates all real clusters from null clusters. The KL divergence ratio is elevated for real clusters overall, but null clusters achieve comparable ratios. Causal steering is positive for most qualified real clusters, but one null cluster (512_138, 0.358) is among the strongest in the dataset. The one measure that does cleanly discriminate is the barycentric predictive advantage: 3 real clusters achieve significant advantages ($p < 10^{-14}$) while zero null clusters do.

The fragmentation of evidence across clusters—768_596 has the best steering and barycentric advantage but murky semantics; 768_140 has the best KL and interpretability but no barycentric advantage; 512_17 has the clearest semantics but weak quantitative validation—is itself informative. It suggests that the discovered structures are not all the same kind of object: some clusters may track discrete, interpretable distinctions (768_140's reference types) that are already captured by a single strong latent, while others may encode graded mixture states (768_596) where the coordinate information provides genuine marginal signal.

7.2 Comparison to Toy Model Expectations

The toy model establishes that, in a fully controlled setting where the generative process is known, the pipeline correctly recovers the ground-truth simplex geometry from SAE latent space with imperfect but directionally correct reconstruction. The real-model results are qualitatively consistent with the toy model: we find simplex-like structures that are partially functional, semantically coherent, and causally relevant but not as cleanly recoverable as in the controlled case. The toy model results contextualize this: even with perfect ground truth, AANet reconstruction is noisy, so the real-model results are consistent with genuine but imperfect encoding of belief geometries under considerably more complex generative dynamics.

7.3 Limitations

Effect sizes are modest. The best steering score (0.419) is well below 1.0. The barycentric R^2 advantage, while significant, corresponds to modest absolute improvements. This is consistent with the simplex structures being present but fragile or partially encoded.

Null cluster size mismatch. Our null clusters contain 47–51 latents compared to 4–26 for real clusters. Larger clusters provide more activation signal, potentially inflating baseline measures. A more controlled comparison would match null clusters to real cluster sizes.

Phantom vertices. Several clusters—real and null—have one poorly-defined vertex with document classification accuracy below 0.15. Steering and classification results involving phantom vertices are unreliable. We note and exclude these where possible but cannot fully eliminate the confound.

No causal proof of belief-state tracking. We cannot directly observe what internal computation Gemma-2-9B performs, and our evidence is correlational (KL, barycentric R^2) or weakly causal (steering, which is a perturbation experiment but not a controlled identification of causal mechanisms). The findings are consistent with belief-state tracking but do not rule out alternative explanations.

Single model, single layer. All real-model results are from Gemma-2-9B, layer 20. Whether the findings generalize to other models, architectures, or layers is unknown.

Degeneracy at large steering scales. At scale $s = 20$, many clusters produce high-degeneracy continuations (model outputs non-text), indicating the model's representations are disrupted before exhibiting clean state transitions. This bounds the strength of causal evidence we can obtain.

8 Conclusion

We have introduced and validated a pipeline for finding simplex-structured representational geometries in SAE latent spaces of language models, motivated by the hypothesis that language models track latent generative states as internal belief states. In a controlled toy model setting, the pipeline correctly recovers known simplex structures from the SAE latent space. Applied to Gemma-2-9B, we identify 13 priority clusters exhibiting $K = 3$ simplex structure, of which 3 show a significant predictive advantage of barycentric over individual latent coordinates—a signal not present in null clusters. Causal steering confirms that AANet-derived simplex coordinates are causally connected to model behavior in 8 of 13 clusters, and all clusters produce semantically coherent and stable vertex labels.

The strongest individual cluster, 768_140, represents the linguistic distinction between generic, specific, and anaphoric reference—a semantically natural candidate for a latent generative state tracked across context. The functionally strongest cluster, 768_596, exhibits both barycentric predictive advantage and the highest causal steering score in the dataset. Together, these findings suggest that Gemma-2-9B's SAE latent space contains subspaces that, at a minimum, encode mixture-like representations of functionally distinct discourse modes—representations that behave in ways expected of belief-state encodings, even if their precise computational role remains to be established.

Future work should address the limitations identified above, including better-matched null clusters, temporal analysis of barycentric coordinates across context windows (to test whether they update in ways consistent with Bayesian belief updating), and extension to additional models and layers.

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A Toy Model Details

Mess3 process. A Mess3 HMM [Marzen and Crutchfield, 2017] has three hidden states $\{0, 1, 2\}$ with transition probabilities parameterized by (x, a): the self-transition probability is a, and the remaining probability $1 - a$ is split uniformly among the other two states. The emission from state s is token s with probability x and a random token with probability $1 - x$.

Tom Quantum (Bloch Walk) process. The Bloch Walk GHMM [Riechers et al., 2025b] is parameterized by (α, β) controlling the step distribution on the Bloch sphere. The internal state is a unit vector on S^2; emissions are four tokens corresponding to projective measurements. The resulting temporal correlations exhibit non-Markovian structure.

Multipartite model vocabulary. The MP model generates tuples from $\{0, \ldots, 3\}^2 \times \{0, 1, 2\}^3$ (two Bloch Walk components $\times$ three Mess3 components), encoded as a single integer token from $\{0, \ldots, 431\}$.

B SAE Details

We use the Gemma-2-9B SAE from the SAELens library (release `gemma-scope-9b-pt-res`, layer 20, expansion factor ≈ 36.6, $K = 50$ active features per token). The SAE was trained by DeepMind on a proprietary mixture of text data using TopK activation with L_2 reconstruction loss; full details are described in the associated technical report.

For the toy model SAEs, we use a custom TopK SAE implementation with $d_{\text{SAE}} = 256$, training for 20,000 steps on sequences generated online from the MP model. We tried $K \in \{3, 4, 5, 6, 7\}$; the best-recovering SAE for the Mess3 component uses $K = 3$.

C AANet Architecture and Training

We use the AANet implementation from Korem et al. [2019]. The encoder consists of two hidden layers (widths 256, 128 by default), a bottleneck of dimension $K-1$ (to enforce the simplex constraint via softmax normalization), and an analogous decoder. The loss combines L_2 reconstruction, a simplex penalty $\lambda_1(1-\|\mathbf{E}(a)\|_1)$, and a non-negativity penalty $\lambda_2 \sum_i |x_i| \mathbf{1}(x_i < 0)$. We train with AdamW, learning rate 10^{-3}, for 10,000 steps, selecting the checkpoint with the lowest validation reconstruction loss. We run 5 independent restarts and average the barycentric coordinates.

D Causal Steering Intervention Types

We use three types of steering interventions, applied at the residual stream at the trigger token position:

- **Type 1**: Ablate and replace. Remove the current cluster activation contribution (subtract the projection onto the cluster subspace) and add back the cluster activation corresponding to the target vertex decoder output.
- **Type 2**: Add direction. Add the difference of AANet decoder outputs at target vs. source vertex barycentric coordinates, scaled by s.
- **Type 3**: Add decoder direction. Add the raw AANet decoder output for the target vertex barycentric coordinate, scaled by s, without subtracting the source vertex contribution.

E All Priority Cluster Results

Table 5: **Full results across all four validation strands for the 13 priority clusters and 3 null clusters.** KL ratio: cross/same symmetric KL divergence ratio. Bary wins: fraction of top-50 tokens where $R^2_{\text{bary}} > R^2_{\text{best latent}}$ ("sig." if Wilcoxon $p < 0.001$). Steering: best-combination steering score ("—" if cluster did not qualify). Interp.: consistency of frontier-LLM vertex labels across 20 iterations (HIGH/MED/LOW).

Cluster	m	KL ratio	Bary wins	Steering	Interp.
512_17	15	1.054	0.00	0.118	HIGH
512_22	6	1.058	0.00	0.356	MED
512_67	25	1.229	0.00	0.250	MED
512_181	26	1.037	1.00 (sig.)	—	HIGH
512_229	22	1.166	1.00 (sig.)	—	MED
512_261	18	1.032	0.00	0.242	MED
512_471	10	1.017	0.00	—	MED
512_504	22	1.164	0.00	—	MED
768_140	26	**1.987**	0.20	0.289	MED
768_210	4	1.017	0.00	0.336	MED
768_306	5	1.102	0.16	—	MED
768_581	11	1.144	0.00	0.209	MED
768_596	6	1.037	0.98 (sig.)	**0.419**	MED
512_138 (null)	47	1.329	0.00	0.358	—
512_345 (null)	51	1.311	0.02	0.150	—
768_310 (null)	48	1.108	0.16	0.272	—

figures/toy_pca_layer2.png

Figure 1: **Simplex structure in the toy model residual stream (layer 2).** Each point is a token position from a held-out sequence. Colors encode the true belief state: RGB channels are proportional to the probability assigned to each of the three hidden states of the Mess3 component (pure red, green, blue at the vertices; mixed colors in the interior). *Top row:* ground-truth belief states. *Bottom row:* belief states predicted by linear readout from the layer-2 residual stream. The triangular (2-simplex) structure of the Mess3 components and the circular (1-simplex product) structure of the Tom Quantum components are clearly visible, confirming that the transformer has learned to represent these belief geometries.

figures/toy_epdf_layer2_k3.png

Figure 2: **Per-latent activation profiles for the toy model (layer 2, TopK $K = 3$).** Each panel shows the activation of a single SAE latent as a function of the ground-truth belief state (heatmap over Mess3 (x, a) parameter space). Latents are sorted by total activation weight. Individual latents fire preferentially near specific vertices or edges of the belief simplex, tiling the simplex geometry in a graded, partially overlapping fashion.

figures/pipeline_diagram.pdf

Figure 3: **Overview of the belief geometry discovery pipeline.** Text sequences are processed through Gemma-2-9B; residual stream activations at layer 20 are encoded by a TopK SAE. Latent decoder directions are partitioned by k-subspace clustering, and candidate $K = 3$ clusters are fit with AANet to recover barycentric (simplex) coordinates. The resulting cluster representations are evaluated with four complementary validation measurements.

figures/cluster_768_596_all_latents.png

(a) Latent space scatter (all samples)

figures/cluster_768_596_centroid_scatter.png

(b) Per-latent centroid positions

Figure 4: **Cluster 768_596: spatial structure.** *(a)* All 5000 sample positions in barycentric coordinates (ternary plot). Concentration near the three vertices indicates that the model visits near-pure vertex states, not just the simplex interior. *(b)* Centroid positions of the 6 individual SAE latents in the cluster. Latents partition across the three vertices, with each vertex attracting one or two latents (centroids near the corresponding corner), consistent with vertex-specialized feature coding.

figures/cluster_768_140_all_latents.png

(a) Latent space scatter (all samples)

figures/768_140_kl_plot.png

(b) KL divergence distributions

Figure 5: **Cluster 768_140: spatial structure and KL divergence.** *(a)* Sample scatter in barycentric coordinates. *(b)* Pairwise symmetric KL divergence distributions for same-vertex (blue), cross-vertex (red), and within-simplex (gray) pairs. The cross-vertex distribution is shifted to substantially higher values, with a cross/same mean ratio of 1.99—by far the strongest KL signal in our dataset. This indicates that tokens near different simplex vertices predict markedly different next-token distributions.

figures/cluster_512_17_vertex_latent_pies.png

(a) Per-vertex latent activation profiles

figures/cluster_512_17_centroid_scatter.png

(b) Per-latent centroid positions

Figure 6: **Cluster 512_17: latent decomposition.** *(a)* Fraction of total cluster activation contributed by each latent at each of the three vertices. Different latents dominate at each vertex, consistent with vertex-specialized feature encoding. *(b)* Centroid positions of the 15 SAE latents. Clear separation across the three simplex vertices confirms that the cluster's latents have distinct vertex preferences.